

Same-row dies for $\lambda = (4, 2, 1^q)$: closing the gap in the $\dim B = 3$ proof

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Abstract

The May-12 evening-3 paper (2026-05-12-evening-4-2-1n-dim3.tex) and the May-13 Shift Lemma paper (2026-05-13-shift-lemma-all-hooks.tex) jointly assert that $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$ and $B_{n-3}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}$ for $\lambda = (4, 2, 1^{n-6})$ at every $n \geq 10$. The argument hinges on a Phase A decomposition $E_{n-1}^+|_{V_\lambda} = \bigoplus_X \Phi_X(V_{\mu_X})$ through three corner pairs $X \in \{A, B, C\}$. This decomposition is *incomplete* for $(4, 2, 1^{n-6})$: the shape has a same-row pair at row 1 (cells $\{(1, 3), (1, 4)\}$), and the corresponding same-row $+q$ -eigenvectors span a non-trivial subspace $S \subset E_{n-1}^+|_{V_\lambda}$ of dimension $f^{(2, 2, 1^{n-6})}$, which the $\bigoplus_X \Phi_X$ decomposition omits.

We close the gap by proving the missing piece: S dies under the iteration $R'_{n-3} \cdots R'_{n-1}$, so the upper bound $\dim \leq 3$ and the E_1^- -containment are recovered.

Theorem (main). For every $q \geq 2$ (equivalently $n \geq 8$), let $\lambda = (4, 2, 1^q)$, $n = q + 6$, $\ell = q + 2$, $j_0 = \ell + 1 = q + 3$, and let $S \subset V_\lambda$ be the span of seminormal basis vectors v_T for SYTs T of λ with $T(1, 3) = n - 1$, $T(1, 4) = n$. Then

$$R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} S = 0.$$

Corollary. The May-12 evening-3 paper's results ($\dim B_{n-3}^{(\lambda)} V_\lambda = 3$ and the E_1^- containment) are now unconditional for every $n \geq 10$. Equivalently, Corollary 4.4 of the Shift Lemma paper, which inherited the gap, is fully closed.

Proof method. The proof is a verbatim adaptation of the May-13 night same-row-dies proof for $(3, 3, 1^q)$, with three ingredients:

1. the $H_q(S_{n-2})$ -equivariant iso $\Psi : S \xrightarrow{\sim} V_{\nu'}$ where $\nu' = (2, 2, 1^q) = \lambda \setminus \{(1, 3), (1, 4)\}$,
2. the Shift Lemma of the May-13 hooks paper (operator identity in $H_q(S_n)$), and
3. the May-13 structural j -formula on $\nu' = (2, 2, 1^q)$ (file 2026-05-13-non-hook-22-1n.tex).

The Shift Lemma rearranges the iteration $R'_{j_0} \cdots R'_{n-1}$ into outer T -factors times a sub-iteration that, on S , equals one R' -step beyond the sharp j -formula iteration on ν' . The trailing $(T_1 + 1)$ of that extra R' -step annihilates the E_1^- that the ν' - j -formula guarantees.

As a by-product we record the corrected Phase A decomposition for any shape with same-row pairs, and the resulting general same-row-dies framework (Theorem 13): for any λ with a same-row pair at row i and $\lambda_i \geq 3$ (so that $\ell(\nu_i) = \ell(\lambda)$), same-row dies for λ at row i as soon as the j -formula on ν_i at the sharp index is known.

1 Setup

Notation follows the May-13 Shift Lemma paper (2026-05-13-shift-lemma-all-hooks.tex) and the May-13 r1 paper (2026-05-13-r1-inductive-reduction.tex). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$. V_λ is the seminormal cell module for $\lambda \vdash n$ in

Hoefsmit's asymmetric Murphy normalization ($b' = 1$). $E_j^\pm := E_j^\pm(V_\lambda)$ denote the q - and (-1) -eigenspaces of T_j . For $1 \leq k \leq n$,

$$R'_k := (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1) \in H_q(S_n),$$

with $R'_1 = 1$. For $1 \leq a \leq b \leq n$, $P_{a,b} := R'_a R'_{a+1} \cdots R'_b$, $P_{a,b} := 1$ for $a > b$. The j -image is

$$B_j^{(\lambda)} V_\lambda := P_{j,n} V_\lambda = R'_j R'_{j+1} \cdots R'_n V_\lambda, \quad j_0(\lambda) := \ell(\lambda) + 1.$$

Throughout this paper. $\lambda = (4, 2, 1^q)$ with $q \geq 2$ (so $n = q + 6 \geq 8$), $\ell := \ell(\lambda) = q + 2$, $j_0 := j_0(\lambda) = q + 3 = n - 3$.

Removable corners of λ .

$c_1 = (1, 4)$ (length-preserving), $c_2 = (2, 2)$ (length-preserving), $c_3 = (q+2, 1)$ (column-1 bottom, length-re

Same-row pair of λ . $\lambda_1 = 4$ and $\lambda_2 = 2 \leq 4 - 2 = \lambda_1 - 2$, so row 1 admits a same-row pair at cells $\{(1, 3), (1, 4)\}$. Row 2 has $\lambda_2 = 2$ but $\lambda_3 = 1 > \lambda_2 - 2 = 0$, so no same-row pair at row 2. Rows ≥ 3 have length 1, no same-row pairs.

The sub-shape.

$$\nu' := \lambda \setminus \{(1, 3), (1, 4)\} = (2, 2, 1^q)$$

on $n - 2 = q + 4$ letters. Note $\ell(\nu') = q + 2 = \ell$ and $j_0(\nu') = q + 3 = j_0$.

Definition 1 (Same-row part S). $S \subseteq V_\lambda$ is the span of $\{v_T : T \in \text{SYT}(\lambda), T(1, 3) = n - 1, T(1, 4) = n\}$. The remaining letters $\{1, \dots, n - 2\}$ fill the cells of ν' , so $\dim S = f^{\nu'}$.

1.1 The $H_q(S_{n-2})$ -iso Ψ

Lemma 2 (Same-row iso). *The map*

$$\Psi : S \xrightarrow{\sim} V_{\nu'}, \quad \Psi(v_T) := v_{T|_{\nu'}}$$

(restriction of T to the letters $\{1, \dots, n - 2\}$ filling the cells of ν') is a $\mathbb{Q}(q)$ -linear isomorphism, equivariant under the subalgebra $\langle T_1, \dots, T_{n-3} \rangle \subset H_q(S_n)$.

Proof. For $T \in S$, the letters $n - 1, n$ occupy $(1, 3), (1, 4) \notin \nu'$, and the letters $\{1, \dots, n - 2\}$ fill the cells of ν' in a way that is row- and column-increasing (since T is), so $T|_{\nu'}$ is a valid SYT of ν' . Conversely, given any SYT T' of ν' , append $n - 1$ at $(1, 3)$ and n at $(1, 4)$ to obtain an SYT $T \in S$ with $T|_{\nu'} = T'$ (the appended cells are at the end of row 1 of λ and the entries are the two largest, so both monotonicity conditions hold). Hence Ψ is a bijection of basis vectors and a $\mathbb{Q}(q)$ -linear iso.

Equivariance. For $1 \leq j \leq n - 3$, T_j acts on letters $j, j + 1$. Both are $\leq n - 2$ and so reside in cells of ν' . The Hoefsmit action on v_T depends only on the axial distance $d_j(T) := c(j + 1) - c(j)$ where $c(r, k) := k - r$. Since $j, j + 1$ occupy the same cells in T and $T|_{\nu'}$, $d_j(T) = d_j(T|_{\nu'})$, and the seminormal-basis action of T_j on v_T matches that on $v_{T|_{\nu'}}$ entry-by-entry (the swap $s_j T$ stays in S iff $s_j T|_{\nu'}$ stays in $\text{SYT}(\nu')$, because the swap only touches cells of ν'). Therefore $T_j \Psi^{-1}(v) = \Psi^{-1}(T_j v)$ for all $v \in V_{\nu'}$, i.e., Ψ is T_j -equivariant. $\square$

Remark 3. Since Ψ is T_1 -equivariant (as $1 \leq n - 3$ for $n \geq 4$), Ψ identifies $E_1^-|_S$ with $E_1^-|_{V_{\nu'}}$. Hence $\Psi^{-1}(E_1^-|_{V_{\nu'}}) = S \cap E_1^-|_{V_\lambda}$.

1.2 The j -formula on $\nu' = (2, 2, 1^q)$

Theorem 4 (May-13 j -formula for $(2, 2, 1^q)$; 2026-05-13-non-hook-22-1n.tex, Theorem 1). *For every $m \geq 6$ (equivalently $q \geq 2$), $\nu' = (2, 2, 1^q) \vdash m = q + 4$ satisfies*

$$B_{m-1}^{(\nu')} V_{\nu'} = R_{m-1}' R_m^{(\nu')} V_{\nu'} \subseteq E_1^- |_{V_{\nu'}}.$$

Remark 5. $\ell(\nu') = q + 2$ and $|\nu'| = q + 4$, so the sharp index is $j_0(\nu') = q + 3 = m - 1$. The May-13 paper's $R_{n-1}' R_n'$ (in their notation) is exactly $B_{j_0(\nu')}^{(\nu')} V_{\nu'}$ in our notation, on $m = n_{\text{paper}} = q + 4$ letters.

2 The Shift Lemma and the main proof

Lemma 6 (Shift Lemma; May-13 Shift Lemma paper, Lemma 1). *For $2 \leq a \leq b \leq n$, inside $H_q(S_n)$:*

$$P_{a,b} = (T_{a-1} + 1)(T_a + 1) \cdots (T_{b-1} + 1) \cdot P_{a-1,b-1}.$$

Theorem 7 (Same-row dies for $\lambda = (4, 2, 1^q)$, $q \geq 2$). *With λ, ℓ, j_0, S as in Section 1, for every $q \geq 2$,*

$$R_{j_0}' R_{j_0+1}' \cdots R_{n-1}' S = 0. \quad (1)$$

Proof. The iteration in (1) has indices $j_0, j_0 + 1, \dots, n - 1$. Since $j_0 = n - 3$, this is exactly three R' -applications: $R_{n-3}' R_{n-2}' R_{n-1}' = P_{n-3,n-1}$.

Step 1: Shift Lemma. Apply Lemma 6 with $a = n - 3 = j_0$ and $b = n - 1$:

$$P_{n-3,n-1} = (T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1) P_{n-4,n-2}. \quad (2)$$

Note $n - 4 = q + 2 = \ell$, so the inner factor is $P_{\ell,n-2}$.

Step 2: $P_{\ell,n-2} S = 0$ via Ψ and the ν' - j -formula. The operator $P_{\ell,n-2} = R_{\ell}' R_{\ell+1}' \cdots R_{n-2}'$ is a product of factors $(T_j + 1)$ for $j \leq n - 3$. Each such factor preserves S as a vector space (since T_j for $j \leq n - 3$ does not move letters $n - 1, n$ out of their cells $(1, 3), (1, 4)$). By Lemma 2, Ψ intertwines T_j -action on S with that on $V_{\nu'}$ for $j \leq n - 3$, hence

$$\Psi(P_{\ell,n-2} |_S \cdot v) = P_{\ell,n-2}^{(\nu')} \Psi(v) \quad \text{for every } v \in S. \quad (3)$$

On the right-hand side, $P_{\ell,n-2}^{(\nu')} = R_{\ell}^{(\nu')} R_{\ell+1}^{(\nu')} \cdots R_{n-2}^{(\nu')}$. Since $\ell(\nu') = q + 2 = \ell$ and $|\nu'| = q + 4 = n - 2$, this factors as

$$P_{\ell,n-2}^{(\nu')} = R_{\ell}^{(\nu')} \cdot \underbrace{R_{\ell(\nu')+1}^{(\nu')} \cdots R_{|\nu'|}^{(\nu')}}_{= B_{j_0(\nu')}^{(\nu')} \text{ (as an operator on } V_{\nu'})}.$$

By Theorem 4 applied to ν' at $m = q + 4 \geq 6$,

$$B_{j_0(\nu')}^{(\nu')} V_{\nu'} \subseteq E_1^- |_{V_{\nu'}}.$$

The operator $R_{\ell(\nu')}^{(\nu')} = (T_{\ell(\nu')-1} + 1) \cdots (T_2 + 1)(T_1 + 1)$ has rightmost factor $(T_1 + 1)$ (applied first to any input vector), and $(T_1 + 1)$ annihilates $E_1^- |_{V_{\nu'}}$. Therefore

$$R_{\ell(\nu')}^{(\nu')} \cdot B_{j_0(\nu')}^{(\nu')} V_{\nu'} \subseteq R_{\ell(\nu')}^{(\nu')} \cdot E_1^- |_{V_{\nu'}} = 0.$$

Hence $P_{\ell, n-2}^{(\nu')} V_{\nu'} = 0$, so by (3) and the injectivity of Ψ ,

$$P_{\ell, n-2} S = \Psi^{-1}(P_{\ell, n-2}^{(\nu')} V_{\nu'}) = \Psi^{-1}(0) = 0.$$

Conclusion. Substituting into (2),

$$P_{n-3, n-1} S = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) P_{\ell, n-2} S = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \cdot 0 = 0. \quad \square$$

3 Closing the gap in the dim formula

We recall the May-12 evening-3 / Shift Lemma argument and identify precisely where Theorem 7 is needed.

Lemma 8 (Corrected Phase A decomposition for λ). *For $\lambda = (4, 2, 1^q)$ at $q \geq 2$,*

$$E_{n-1}^+ |_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S, \quad (4)$$

where

- $\mu_A = \lambda \setminus \{c_1, c_3\} = (3, 2, 1^{q-1})$ on $n-2$ letters (the May-12 today family); $\Phi_A : V_{\mu_A} \hookrightarrow V_\lambda$ is the $+q$ -eigenvector embedding of the T_{n-1} -2-block at the corner pair $\{c_1, c_3\}$.
- $\mu_B = \lambda \setminus \{c_2, c_3\} = (4, 1^q)$ (hook on $n-2$ letters); Φ_B analogously at the corner pair $\{c_2, c_3\}$.
- $\mu_C = \lambda \setminus \{c_1, c_2\} = (3, 1^{q+1})$ (hook on $n-2$ letters); Φ_C analogously at the corner pair $\{c_1, c_2\}$.
- S as in Definition 1.

Proof. The May-20 paper's Lemma 4.4 (Phase A endpoint) gives $R'_n V_\lambda = E_{n-1}^+ |_{V_\lambda}$ for any λ . Every $T_{n-1} +q$ -eigenvector in the seminormal basis arises from one of two configurations of the letters $\{n-1, n\}$ in an SYT T of λ :

- (1) $\{n-1, n\}$ at a corner pair $\{c, c'\}$ in different rows and columns: T_{n-1} is a non-degenerate 2×2 Hoefsmit block on $\text{span}(v_T, v_{s_{n-1}T})$ with one-dim $+q$ -eigenvector $\Phi_{\{c, c'\}}(v_T|_{\mu_{\{c, c'\}}})$. The three corner pairs of λ are $\{c_1, c_3\} = A$, $\{c_2, c_3\} = B$, $\{c_1, c_2\} = C$.
- (2) $\{n-1, n\}$ at a same-row pair: $T_{n-1}v_T = qv_T$, so v_T itself is a $+q$ -eigenvector. The unique same-row pair of λ is $\{(1, 3), (1, 4)\}$ (row 1).

The supports of the four subspaces in (4) are pairwise disjoint subsets of seminormal basis indices (each subspace is supported on SYTs with $\{n-1, n\}$ at a specific pair of cells), so the sum is direct.

The dimension count: $\dim \Phi_X(V_{\mu_X}) = f^{\mu_X}$ (the Φ_X -image has one $+q$ -eigenvector per 2-block, and each 2-block is in bijection with an SYT $T' \in \text{SYT}(\mu_X)$), and $\dim S = f^{\nu'}$. The $+q$ -eigenvectors of T_{n-1} exhaust this enumeration, so the four-piece sum equals E_{n-1}^+ . $\square$

Remark 9. Lemma 4.3(3) of the May-12 evening-3 paper and Lemma 4.4 of the May-20 paper, as written, both omit the same-row part S from the Phase A decomposition. For $\lambda = (3, 2, 1^{n-5})$ (the May-20 family), λ has *no* same-row pair, so the stated decomposition is correct. For $\lambda = (4, 2, 1^{n-6})$ (the May-12 evening-3 family), the same-row pair at row 1 is present and $\dim S = f^{(2, 2, 1^{n-6})} > 0$. Computationally, $\dim S = 20$ at $n = 10$ and $\dim S = 27$ at $n = 11$ (verified mod 100 003, $Q = 11$).

3.1 The corrected upper bound

Theorem 10 (Upper bound, closing the gap). *For $\lambda = (4, 2, 1^{n-6})$ at every $n \geq 10$,*

$$\dim B_{n-3}^{(\lambda)} V_\lambda \leq 3, \quad \text{and} \quad B_{n-3}^{(\lambda)} V_\lambda \subseteq E_1^-|_{V_\lambda}.$$

Proof. By the corrected Phase A decomposition (Lemma 8),

$$B_{n-3}^{(\lambda)} V_\lambda = P_{n-3,n-1} R'_n V_\lambda = P_{n-3,n-1} (\Phi_A(V_{\mu_A}) \oplus \Phi_B(V_{\mu_B}) \oplus \Phi_C(V_{\mu_C}) \oplus S).$$

Φ_X -pieces, $X \in \{A, B, C\}$. The May-12 evening-3 paper's Theorem 3 (the three-branch reduction) gives, for each X :

$$P_{n-3,n-1} \Phi_X(V_{\mu_X}) = (T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1) \Phi_X(B_{n-4}^{(\mu_X)} V_{\mu_X}).$$

This reduction is unconditional — its proof uses only operator-theoretic commutation through Φ_X (index-gap arguments and equivariance under $\langle T_1, \dots, T_{n-3} \rangle$). Then:

- $\dim B_{n-4}^{(\mu_A)} V_{\mu_A} = 2$ by the May-12 today theorem on $(3, 2, 1^{m-5})$ at $m = n - 2 \geq 8$.
- $\dim B_{n-4}^{(\mu_B)} V_{\mu_B} = 1$ by the Shift Lemma paper's main theorem (hook j -formula at $k = 4$) and the corresponding $\dim \leq 1$ corollary, applied to $\mu_B = (4, 1^{m-4})$ at $m = n - 2 \geq 8$.
- $B_{n-4}^{(\mu_C)} V_{\mu_C} = 0$ by the j -formula leakage on $\mu_C = (3, 1^{m-3})$: the hook j -formula at $k = 3$ gives $B_{m-1}^{(\mu_C)} V_{\mu_C} \subseteq E_1^-$, and applying one further $R'_{n-4}^{(\mu_C)}$ ends with $(T_1 + 1)$ annihilating E_1^- .

Total contribution: $\leq 2 + 1 + 0 = 3$, and all in $E_1^-|_{V_\lambda}$ (since each μ_X -image is in $E_1^-|_{V_{\mu_X}}$, Φ_X preserves E_1^- , and the outer T -factors $(T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)$ have indices $\geq n - 4 = q + 2 \geq 4$, hence commute with T_1 and preserve E_1^-).

S-piece. By Theorem 7, $P_{n-3,n-1} S = 0$. This contributes 0 to the total — which is precisely the contribution the gapped argument silently assumed without proof.

Sum. $\dim B_{n-3}^{(\lambda)} V_\lambda \leq 2 + 1 + 0 + 0 = 3$, and the image lies in $E_1^-|_{V_\lambda}$. $\square$

Corollary 11 ($\dim = 3$ unconditional). *For $\lambda = (4, 2, 1^{n-6})$ at every $n \geq 10$, $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$.*

Proof. The upper bound is Theorem 10. The matching lower bound is the May-12 evening-3 paper's Proposition 9 (explicit linear independence of $F_{A,A}, F_{A,B}, F_B$ via position-of- n and position-of- $(n-1)$ separators). The lower-bound argument is independent of the Phase A decomposition — it constructs three explicit vectors in $B_{n-3}^{(\lambda)} V_\lambda$ and separates them by projections — and so is unaffected by the gap. $\square$

Corollary 12 (Cor. 4.4 of Shift Lemma paper, unconditional). *For $\lambda = (4, 2, 1^{n-6})$ at every $n \geq 10$, $\Pi^{S_n}|_{V_\lambda} = 0$ (rank-zero).*

Proof. By Theorem 10, $B_{n-3}^{(\lambda)} V_\lambda \subseteq E_1^-$. Iterating: $B_{n-3}^{(\lambda)} V_\lambda = R'_{n-4} B_{n-3}^{(\lambda)} V_\lambda \subseteq R'_{n-4} E_1^- = 0$, since R'_{n-4} ends with $(T_1 + 1)$ on the right. Successive applications of R'_j for $2 \leq j \leq n - 5$ each preserve 0, so $\Pi^{S_n}|_{V_\lambda} = B_2^{(\lambda)} = 0$. $\square$

4 The general same-row-dies framework

The proof of Theorem 7 factors into three ingredients that do not depend on the specific choice $\lambda = (4, 2, 1^q)$. We isolate the general framework.

Theorem 13 (General same-row dies). *Let $\lambda \vdash n$, and let i be a row of λ admitting a same-row pair, i.e., $\lambda_i \geq 2$ and $\lambda_{i+1} \leq \lambda_i - 2$. Set*

$$\nu_i := \lambda \setminus \{(i, \lambda_i - 1), (i, \lambda_i)\}$$

on $n-2$ letters, and let $S_i \subset V_\lambda$ be the span of seminormal basis vectors v_T with $T(i, \lambda_i - 1) = n-1$, $T(i, \lambda_i) = n$. Set $\ell := \ell(\lambda)$, $j_0 := j_0(\lambda) = \ell + 1$. Assume:

(A) $\ell(\nu_i) = \ell$ (equivalently, the same-row pair is not the only cells of row i , i.e., $\lambda_i \geq 3$, OR $i < \ell$, $\lambda_i = 2$, and $\lambda_{i+1} = 0$ — the latter impossible because $\lambda_{i+1} \geq 1$ when $i < \ell$; hence simply $\lambda_i \geq 3$).

(B) $B_{j_0(\nu_i)}^{(\nu_i)} V_{\nu_i} \subseteq E_1^-|_{V_{\nu_i}}$ (the j -formula on ν_i at its sharp index).

Then

$$P_{j_0, n-1} S_i = R'_{j_0} R'_{j_0+1} \cdots R'_{n-1} S_i = 0.$$

Proof. Mirror the proof of Theorem 7. Condition (A) ensures that $j_0(\nu_i) = \ell(\nu_i) + 1 = \ell + 1 = j_0$ (same sharp index as λ), and that the Shift Lemma's inner factor $P_{\ell, n-2}$ on S_i , via the same-row iso $\Psi_i : S_i \xrightarrow{\sim} V_{\nu_i}$, corresponds to $P_{\ell, n-2}^{(\nu_i)} = R_{\ell}^{(\nu_i)} \cdot B_{j_0(\nu_i)}^{(\nu_i)} V_{\nu_i}$. By (B), $B_{j_0(\nu_i)}^{(\nu_i)} V_{\nu_i} \subseteq E_1^-|_{V_{\nu_i}}$, and the trailing $(T_1 + 1)$ of $R_{\ell}^{(\nu_i)}$ annihilates this. The Shift Lemma's outer factors $(T_{\ell} + 1) \cdots (T_{n-2} + 1)$ contribute zero on a zero argument. Hence $P_{j_0, n-1} S_i = 0$.

The iso Ψ_i is constructed exactly as in Lemma 2: $\Psi_i(v_T) := v_{T|_{\nu_i}}$, bijective and $\langle T_1, \dots, T_{n-3} \rangle$ -equivariant under the same axial-distance argument. $\square$

Remark 14 (Reach of the framework). Theorem 13 subsumes:

- Hooks $\lambda = (k, 1^{n-k})$ at $k \geq 3$ (row 1 same-row pair, $\nu_1 = (k-2, 1^{n-k})$ a hook with smaller arm). Done by the Shift Lemma paper (May-13).
- $\lambda = (3, 3, 1^q)$ (row 2 same-row pair, $\nu_2 = (3, 1^{q+1})$ a hook). Done by the same-row-dies-331 paper (May-13 night).
- $\lambda = (4, 2, 1^q)$ at every $q \geq 2$ (row 1 same-row pair, $\nu_1 = (2, 2, 1^q)$, j -formula by May-13 on $(2, 2, 1^*)$). *This paper.*
- $\lambda = (k, 2, 1^q)$ at $k \geq 5$: row 1 same-row pair, $\nu_1 = (k-2, 2, 1^q)$. Requires j -formula on $(k-2, 2, 1^q)$; recursion is well-founded but waits on the $(k-2, 2, 1^q)$ result.
- $\lambda = (k, k, 1^q)$ at $k \geq 4$: row 2 same-row pair if $k \geq 4$ (since $\lambda_2 = k \geq 4 \geq 2$, $\lambda_3 = 1 \leq k-2$), $\nu_2 = (k, k-2, 1^q)$. Requires j -formula on $(k, k-2, 1^q)$. Open in general.

5 Computational verification

Scripts: `~/projects/scratch/2026-05-13-same-row-dies/track_S421.py`. Mod $P = 100\,003$, $Q = 11$.

Phase A decomposition (Lemma 8). At $n = 10, 11$:

n	$\dim V_\lambda$	$\dim \Phi_A$	$\dim \Phi_B$	$\dim \Phi_C$	$\dim S$	$\dim E_+ _{V_\lambda}$
10	350	64	35	21	20	140
11	594	105	56	28	27	216

In both cases, $\dim \Phi_A + \dim \Phi_B + \dim \Phi_C + \dim S = \dim E_+$, and the 4-piece direct sum is verified. The omission of S from the May-12 evening-3 / May-20 statements is real: at $n = 10$, $\dim(\Phi_A + \Phi_B + \Phi_C) = 120 \neq 140 = \dim E_+$, missing $20 = \dim S$.

Same-row dies (Theorem 7). Applying R'_{n-1} , R'_{n-2} , R'_{n-3} successively to S :

n	$\dim S$	after R'_{n-1}	after R'_{n-2}	after R'_{n-3}
10	20	5	1	0
11	27	6	1	0

The decay pattern $\dim S \rightarrow (q+1) \rightarrow 1 \rightarrow 0$ matches the prediction from the $\nu' = (2, 2, 1^q)$ -iteration: $\dim V_{\nu'} = f^{(2,2,1^q)} \rightarrow \dim E_+|_{V_{\nu'}} \rightarrow \dim B_{j_0(\nu')}^{(\nu')} V_{\nu'} \rightarrow 0$, with the last step the May-13 $(2, 2, 1^q)$ - j -formula's E_1^- containment plus the trailing $(T_1 + 1)$.

Final image $\dim B_{n-3}^{(\lambda)} V_\lambda = 3$. Verified at $n \in \{10, 11\}$ via direct rank computation (already in the May-12 evening-3 paper), independently of this paper's contribution.

6 Honest status

What we proved. Theorem 7 (same-row dies for $(4, 2, 1^q)$ at every $q \geq 2$, structurally), and the general framework Theorem 13. As corollaries: Corollary 11 ($\dim = 3$ unconditional for $(4, 2, 1^{n-6})$ at $n \geq 10$) and Corollary 12 (rank-zero for the same family). These had been asserted by the May-12 evening-3 paper and the May-13 Shift Lemma paper, but the proofs as written had a Phase A decomposition gap; this paper fills it.

What we did not prove. The lower-bound argument (Proposition 9 of May-12 evening-3) is unaffected by the gap and is reused without modification. The Catalan-support description of the $\mu_B = (4, 1^{m-4})$ -hook (May-11 Conjecture 5.1, part (ii)) is unrelated and remains open. The j -formula on $(k, 2, 1^q)$ for $k \geq 5$ and on $(k, k-2, 1^q)$ for $k \geq 4$ remain open in general.

What it took. The proof is short — one operator identity (the Shift Lemma, already in hand), one iso ($\Psi : S \xrightarrow{\sim} V_{\nu'}$, by the same axial-distance argument that gave the iso for hooks), and one earlier theorem (May-13 j -formula on $(2, 2, 1^q)$). The contribution is a careful audit of the Phase A decomposition, identification of the missing S -piece, and the verification that the same framework that closes the hook same-row-dies (Shift Lemma + iso + j -formula on the sub-shape) closes this case too, with the sub-shape now being the *non-hook* $(2, 2, 1^q)$.

Relationship to the dim-formula program.

- *Filling the gap* makes the refined recursive rank-corner formula (May-12 evening-3 Conjecture 12.1) unconditional at the smallest test case where it differs from the original May-11 formula. The recursive prediction $\dim B_{j_0}^{(\lambda)} = \dim B^{(\mu_A)} + \dim B^{(\mu_B)} = 2 + 1 = 3$ is now rigorous end-to-end.
- *Framework reach.* Theorem 13 predicts same-row dies for any λ as soon as the j -formula on the sub-shape ν_i at the sharp index is known. Combined with the multi-corner Φ -piece machinery, this should propagate the dim formula through all r -multi-corner shapes *provided* the j -formula on smaller shapes can be inductively established.